

Analysis

PENG

Created: 20260801, Version: 20260805

Contents

1	Numbers	1
1.1	Natural Number system	1
1.1.1	The Peano Axioms	1
1.1.2	Addition	2
1.1.3	Multiplication	3

1 Numbers

1.1 Natural Number system

- This is how mathematics works - it treats its objects abstractly, caring only about what properties the objects have, not what the objects are or what they mean [1].
- It is pointless, at least from a mathematician's standpoint, as to argue about which model is the true one; as long as it obeys all the axioms and does all the right things, that's good enough to do maths.

1.1.1 The Peano Axioms

Axiom 1.1. *Peano axioms:*

1. 0 is a natural number.
2. If n is a natural number, then $n++$ is also a natural number.
3. $n++ \neq 0$
4. $n \neq m \Rightarrow n++ \neq m++$
 $n++ = m++ \Rightarrow n = m$
5. Principle of mathematical induction:
Suppose that $P(0)$ is true, and suppose that whenever $P(n)$ is true, $P(n++)$ is also true. Then $P(n)$ is true for every natural number n .

Definition 1.2.

$$\begin{aligned}1 &:= 0++ \\2 &:= 1++ \\3 &:= 2++ \\&\vdots\end{aligned}$$

1.1.2 Addition

Definition 1.3. *Addition.*

Let n and m be natural numbers.

1. $0 + m := m$
2. $(n++) + m := (n + m)++$

Lemma 1.4. *Let n and m be natural numbers.*

1. $n + 0 = n$
2. $n + (m++) = (n + m)++$

Proposition 1.5. *Let a, b, c be natural numbers.*

1. $a + b = b + a$
2. $(a + b) + c = a + (b + c)$
3. $a + b = a + c \Rightarrow b = c$

Definition 1.6. *A natural number n is positive iff (if and only if) $n \neq 0$.*

Proposition 1.7. *Let a and b be natural numbers.*

1. *If a is positive, then $a + b$ is positive.*
2. $a + b = 0 \Rightarrow a = b = 0$.
3. *If a is positive, then there exists only one natural number b such that $b++ = a$.*

Proof. Prop. 1.7 (1)

$a + 0 = a$ is positive. Suppose that $a + b$ is positive. Then $a + b++ = (a + b)++ \neq 0$ is positive. $\square$

Proof. Prop. 1.7 (3)

Suppose that $a = b++$. Then $a++ = (a)++$. $\square$

Definition 1.8. *Let n and m be natural numbers.*

1. $n \geq m$ or $m \leq n$ iff $n = m + a$
2. $n > m$ or $m < n$ iff $n \geq m$ and $n \neq m$

Proposition 1.9. *Let a, b, c be natural numbers.*

1. $a \geq a$
2. *If $a \geq b$ and $b \geq c$, then $a \geq c$*
3. *If $a \geq b$ and $b \geq a$, then $a = b$*
4. $a \geq b$ iff $a + c \geq b + c$
5. $a < b$ iff $a++ \leq b$
6. $a < b$ iff $b = a + d$ for some positive number d

Proof. Prop. 1.9 (5)

$b = a + d$ and $b \neq a \Rightarrow d \neq 0 \Rightarrow d = n++$
 $b = a + (n++) = (a++) + n \Rightarrow a++ \leq b$ $\square$

Proposition 1.10. *Let a and b be natural numbers. Then exactly one of following statement is true: $a < b$, $a = b$, or $a > b$.*

Proof. If $a < b$ and $a > b \xrightarrow{\text{Prop. 1.9 (3)}} a = b$ contradiction $\square$

Proposition 1.11. *Strong principle of induction.*

Let m_0 be a natural number. Suppose that for each $m \geq m_0$, if $P(m')$ is true for all $m_0 \leq m' < m$, then $P(m)$ is also true. Then $P(m)$ is true for all $m \geq m_0$.

Proof. Let $Q(n)$ be the property that $P(m)$ is true for all $m_0 \leq m < n$.

Suppose that if $Q(n)$ is true, then $P(m)$ is also true.

Then $P(m)$ is true for all $m_0 \leq m < n++$. $\square$

Example 1.12. *Fibonacci numbers.*

$$F_n = \frac{\phi^n - \psi^n}{\phi - \psi} \quad (1.1)$$

$$F_{n+2} = F_{n+1} + F_n \quad (1.2)$$

where $\phi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$

Exercise 1.13. *Principle of backwards induction.*

Let n be a natural number. Whenever $P(m++)$ is true, $P(m)$ is true. If $P(n)$ is true, then $P(m)$ is true for all $m \leq n$.

Proof. Suppose that if $P(n)$ is true, then $P(m)$ is true for all $m \leq n$. If $P(n++)$ is true, then $P(m)$ is true for all $m \leq n++$. $\square$

Exercise 1.14. *Principle of induction starting from the base case n .*

Let n be a natural number. Whenever $P(m)$ is true, $P(m++)$ is true. If $P(n)$ is true, then $P(m)$ is true for all $m \geq n$.

Proof. If $P(0)$ is true, suppose that $P(m)$ is true. Then $P(m++)$ is also true. Then $P(m)$ is true for all $m \geq 0$. Suppose that if $P(n)$ is true, then $P(m)$ is true for all $m \geq n$. If $P(n++)$ is true, then $P(m)$ is true for all $m \geq n++$. $\square$

1.1.3 Multiplication

Definition 1.15. Let n and m be natural numbers.

1. $0 \times m := 0$
2. $(n++) \times m := (n \times m) + m$

Lemma 1.16. Let n and m be natural numbers.

1. $n \times m = m \times n$
2. $n \times m = 0$ iff at least one of n and m is equal to 0.

Proposition 1.17. Let a, b, c be natural numbers.

1. $a(b+c) = (b+c)a = bc+ca$
2. $(ab)c = a(bc)$
3. If $a < b$ and c is positive, then $ac < bc$.
4. If $ac = bc$ and c is positive, then $a = b$.

Lemma 1.18. *Euclid's division lemma.*

Let n be a natural number and q be a positive natural number. Then there exist natural numbers m and r , such that $0 \leq r < q$ and $n = mq + r$.

Definition 1.19. *Exponentiation.*

1. $m^0 := 1$ and $0^0 := 1$
2. $m^{n++} := m^n \times m$

References

- [1] Terence Tao. *Analysis I*. Vol. 1. Springer, 2022. ISBN: 978-981-19-7261-4.